POWERS & ROOTS

LECTURE

by

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LECTURE OBJECTIVES

- Use rules to operate with powers
- Understand roots and fractional exponents
- Learn Zero exponents
- Learn negative exponents
- Evaluate problems.

OPERATIONS WITH POWERS

Powers enable us to write a repeated operation in a shorthand way

eg $a \times a \times a \times a \times a = a^5$ (a to the power 5)

$a^n = a \times a \times a \times \ldots\ldots a$

(where a appears n times after the $=$ sign)

In a^n: $a =$ base

$n =$ exponent or index or power

ADDITION AND SUBTRACTION

To add or subtract like powers simply add or subtract their coefficients

$$3a^2 + b^3 - a^2 + 4b^3 = 2a^2 + 5b^3$$

$$3a^4 + ba^4 = (3 + b)\,a^4$$

MULTIPLICATION OF POWERS

Powers with different bases cannot be simplified

$$a^2 \times b^3 \neq (ab)^6$$

To multiply powers with the same base, add their indices

$$a^2 \times a^3 = a^5$$

DIVISION OF POWERS

Powers with different bases cannot be simplified

$$a^5 \div b^3 \neq \left(\frac{a}{b}\right)^2$$

To divide powers with the same base subtract the index of the denominator from that of the numerator

$$a^n \div a^m = a^{n-m}$$

RAISING A PRODUCT TO AN EXPONENT

Raise the individual factors to the exponent

$$(4bc)^2 = 4^2 b^2 c^2 = 16b^2 c^2$$

$$(a+b)^2 \neq a^2 + b^2$$

Note $(3+1)^2 = 4^2 = 16$

$$\neq 3^2 + 1^2 = 9 + 1 = 10$$

RAISING A FRACTION TO AN EXPONENT

Raise both the numerator and the denominator to the exponent

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$\left(\frac{6}{2}\right)^2 = \frac{6^2}{2^2} = \frac{36}{4} = 9$$

RAISING A POWER TO AN EXPONENT

Use rule for multiplying powers

$$(a^m)^n = a^{mn}$$

$$(a^2)^3 = a^2 \times a^2 \times a^2 = a^6 = a^{2\times 3}$$

Hence $(5^2)^9 = 5^{(2)9} = 5^{18}$

FRACTIONAL EXPONENTS

A root $\left(\sqrt{\ }\right)$ is shorthand for the inverse

$\left(\frac{1}{n}, \text{where n is a positive integer}\right)$ of

the operation of raising to a power.

Root $= \sqrt{\ }$ = square root i.e. $\sqrt[2]{\ }$

$9^{\frac{1}{2}} = \sqrt{9}$ stands for the operation of finding

the number which gives the result 9

when raised to the power of 2

FRACTIONAL EXPONENTS

The nth root of a quantity, $\left(a^{\frac{1}{n}} \right)$, written

$\sqrt[n]{a}$ is a quantity b such that $b^n = a$

N.B. An even root does not give a unique result

i.e. $\sqrt{64} = \pm 8$

A negative number cannot have an even root

No number can be squared to produce - 4

FRACTIONAL EXPONENTS

Powers where the exponent is

not an integer i.e. $\frac{m}{n}$

$$a^{\frac{m}{n}} = a^{m \times \frac{1}{n}} \text{ or } a^{\frac{1}{n} \times m}$$

$$= \left(a^{m}\right)^{\frac{1}{n}} \text{ or } \left(a^{\frac{1}{n}}\right)^{m}$$

$$= \sqrt[n]{a^{m}} \text{ or } \left(\sqrt[n]{a}\right)^{m}$$

Example

$$\left(a^{m}\right)^{\frac{1}{n}} \text{ or } \left(a^{\frac{1}{n}}\right)^{m}$$

$$4^{\frac{7}{2}} = \left(4^{7}\right)^{\frac{1}{2}} \text{ or } \left(4^{\frac{1}{2}}\right)^{7}$$

$$= \sqrt{4^{7}} \quad \text{or } \left(\sqrt{4}\right)^{7}$$

$$= \sqrt{16384} \qquad \text{or } \left(\pm 2\right)^{7}$$

$$= \pm 128$$

ZERO EXPONENT

Powers where the exponent is zero

Any power with an exponent of 0 equals 1

Note $a^0 = 1$

$$\frac{3^2}{3^2} = 3^{2-2} = 3^0 = 1$$

or $$\frac{9}{9} = 1$$

NEGATIVE EXPONENT

$$a^{-n} = \frac{1}{a^{n}}$$

$$2^{-2} = \frac{1}{2^{2}} = \frac{1}{4}$$

Common Sources of Error in Handling Operations with Powers

Correct

$$(ab)^2 = a^2b^2$$

$$\sqrt{(a^2b^2)} = \sqrt{a^2} \times \sqrt{b^2} = ab$$

$$a^{-2} + b^{-2} = \frac{1}{a^2} + \frac{1}{b^2}$$

Common Sources of Error in Handling Operations with Powers

Incorrect

$$(a+b)^2 = a^2 + b^2$$

$$\sqrt{(a^2 + b^2)} = \sqrt{a^2} + \sqrt{b^2}$$

$$a^{-2} + b^{-2} = \frac{1}{\left(a^2 + b^2\right)}$$

RULES OF EXPONENTS

$a^m \times a^n = a^{m+n}$ for $m,n \in N$

$\frac{a^m}{a^n} = a^{m-n}$ for $m,n \in N, m > n, a \neq 0$

$\left(a^m\right)^n = a^{mn}$ for $m,n \in N$

$a^{-m} = \frac{1}{a^m}$ for $a \in R,\ a \neq 0$

$a^0 = 1$ for $a \in R$, a $\neq 0$ $\left(\text{Note}: 0^0 \text{ is meaningless!}\right)$

RULES OF EXPONENTS

$$a^{\frac{1}{m}} = \sqrt[m]{a} \qquad \text{for } a \in R,\ n \in N$$

$$a^{\frac{n}{m}} = \left(a^{\frac{1}{m}}\right)^n = \left(\sqrt[m]{a}\right)^n$$

$$\text{OR} = \left(a^n\right)^{\frac{1}{m}} = \sqrt[m]{a^n} \quad \text{provided the } m^{th} \text{ root of } a^n \text{ exists}$$

N – Natural numbers (counting numbers)

R - Real numbers (numbers that exist as opposed to complex numbers)

e.g. of real numbers: whole numbers, integers, decimal and vulgar fractions

Indicate if every relationship is true or false:

1. $5^3 = 15$ T F
2. $2^4 = 16$ T F
3. $5^4 \cdot 5^5 = 5^{20}$ T F
4. $(5^3)^7 = 5^{21}$ T F
5. $0^0 = 1$ T F
6. $8^3 \div 8^3 = 0$ T F
7. $(7^3 \div 7^2)^0 = 1$ T F
8. $(14^2 \div 7^2) = (14 \div 7)^2$ T F
9. $(15^3 \div 3^3) = 125$ T F
10. $(2^4)^3 \div (2^4)^2 = 16$ T F
11. $(3^4)^2 \div 3 = 3^9$ T F
12. $2^3 \ 2^5 \div 2^2 \div 2^8 = 2^{-2}$ T F

Calculate the value of these expressions using powers properties:

1. $\left\{8^{22} \div \left[\left(8^{2} \cdot 8 \cdot 8^{3}\right)^{2} \div 8\right]\right\} \div \left(8^{4} \cdot 8\right)^{2}$

2. $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$

3. $10^{\frac{1}{3}n} \times 15^{\frac{1}{2}n} \times 6^{\frac{1}{6}n} \div 45^{\frac{1}{3}n}$